

AI and Games

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Deadline: 31st May 2026

Approved on 22nd Apr 2026 at 08:53.

Basic module details

Month of introduction/change
September

Year of introduction/change
2026

Module title
AI and Games

Module duration
4 weeks

Department for SITS
WMG

Rationale for introduction, revision or discontinuation
This is a new module for the Games Engineering course as an extra elective. This is being introduced for three reasons: one due to uncertainty if the existing electives will be delivered next year, secondly the electives at the moment are not appropriately focused on games, or at the appropriate level, and thirdly this is a module that was eventually planned to be added to the course and delivered by an expanded course team so should be added now.

This module will be delivered by the new TF role which is currently being recruited for.

Affected departments
• WMG

Module leader
Thomas Bashford-Rogers (2270898)

Level
7

Level for SITS
M1

Credit value
15

Module delivery
Blended

Study locations

Location	Primary
University of Warwick Main Campus, Coventry	Yes

Module description

Introductory description
This module introduces the principles and practical applications of artificial intelligence within games. Game AI has historically been a major driver of innovation in interactive systems, combining algorithmic techniques with data-driven approaches to create believable agents, intelligent behaviours, and adaptive game systems. As modern games increasingly integrate machine learning techniques alongside traditional AI approaches, it is important that graduates

understand both the classical foundations and contemporary developments in this field.

The module is structured around two complementary themes: traditional game AI techniques and modern machine learning methods used in games. Traditional techniques include pathfinding algorithms, behaviour trees, and decision systems, which form the backbone of many interactive and real-time gameplay systems. These approaches emphasise deterministic reasoning, efficient algorithms, and the design of robust decision-making systems suitable for real-time environments.

Alongside these foundations, the module introduces key concepts from modern artificial intelligence and machine learning. Students will explore neural networks and their evolution from simple architectures to more advanced models such as convolutional neural networks, recurrent networks, and transformers. The module also examines how these techniques are applied within the games industry, including AI agents, procedural content generation, rendering enhancement and denoising, physics approximation, and generative systems for textures, models, and dialogue.

The emphasis throughout the module is on understanding core principles and applying them in practice. Through this combination of theory and implementation, the module aims to provide students with both conceptual understanding and practical skills relevant to modern game development and interactive AI systems.

Principal module aims

The over-arching aim of this module is to provide students with the knowledge and practical skills required to understand, design, and implement artificial intelligence techniques used in digital games. The module aims to develop both an understanding of traditional game AI systems and modern machine learning approaches, and how these techniques can be applied to interactive environments.

The sub-aims of this module are:

- Develop an understanding of the fundamental algorithms and techniques used in traditional game AI, including pathfinding, decision systems, and behaviour modelling
- Explore the principles behind modern artificial intelligence and machine learning methods, including neural networks and their variants
- Understand how AI techniques are applied in contemporary games, including intelligent agents, procedural systems, rendering enhancement, and generative content
- Gain practical experience implementing AI techniques within game-related contexts
- Develop the ability to critically evaluate different AI approaches and determine their suitability for particular game design or technical problems

Learning outcomes

- Critically analyse the principles and techniques used in traditional game artificial intelligence, including pathfinding, decision systems, and behaviour modelling.
- Evaluate and compare different AI approaches used in games, including rule-based systems and machine learning methods, and justify their suitability for specific design or technical problems.
- Design and implement classical game AI algorithms within an interactive environment, such as pathfinding or behaviour systems for game agents.
- Demonstrate a conceptual understanding of modern machine learning techniques, including neural networks, convolutional networks, recurrent networks, and transformer architectures.
- Develop and apply neural network-based solutions to solve practical problems relevant to games or interactive systems.
- Critically assess the role and impact of modern AI technologies in game development, including applications such as agent behaviour, content generation, rendering enhancement, and large language model integration.

Outline syllabus for publication

Introduction to Artificial Intelligence in Games Search and Pathfinding Decision Making Systems for Game Agents
Foundations of Machine Learning Neural Network Architectures AI Agents and Learning in Games AI for Game
Content and Systems Modern AI Applications in Games

Should this module have a reading list?

No

Module delivery

Minimum student number

20

Maximum student number

30

Estimated student number

90

Teaching split

Department	Percentage
WMG	100%

Skills, competencies, and attributes

Subject specific skills

Mathematical skills, programming skills, AI skills

Transferable skills

Technology literacy, adaptability

Includes research element

No

Interdisciplinary

No

International

No

Learning hours breakdown

Lecture

Required: 15 × 1 hour Optional: 0 × 0 hours

Seminar

Required: 0 × 1 hour Optional: 0 × 0 hours

Tutorial

Required: 15 × 1 hour Optional: 0 × 0 hours

Project Supervision

Required: 0 × 0 hours Optional: 0 × 0 hours

Demonstration

Required: 0 × 0 hours Optional: 0 × 0 hours

Practical class

Required: 0 × 0 hours Optional: 0 × 0 hours

Supervised practical class

Required: 0 × 0 hours Optional: 0 × 0 hours

Fieldwork

Required: 0 × 0 hours Optional: 0 × 0 hours

External visit

Required: 0 × 0 hours Optional: 0 × 0 hours

Work based learning

Required: 0 × 0 hours Optional: 0 × 0 hours

Online learning (scheduled sessions)

Required: 0 × 0 hours Optional: 0 × 0 hours

Online learning (independent)

Required: 20 × 1 hour Optional: 0 × 0 hours

Autonomous skills development (Degree apprenticeship provision only)

Required: 0 × 0 hours Optional: 0 × 0 hours

Other activity

Required: 0 × 1 hour Optional: 0 × 0 hours

Placement

0 hours

Private study

40 hours

Details of private study and independent learning

Private study will include exploration and prototyping of AI algorithms.

Assessment

60 hours

Total study hours for module

150 hours

Assessment

Reassessment model

Individual

Assessment components

Title	Type	Length	Weight	Final chronological	Eligible for self-certification
Game AI Implementation	Project - Individual	4000 words	100%	Yes	Yes (extension)
Student preparation and completion time (hours)	60				
Description	Implementation of AI into a game environment				
Reassessment component					
Title	Type		Length		Eligible for self-certification
Game AI Implementation	Project - Individual		4000 words		No
Description	Implementation of AI into a game environment				

Must the student pass all parts of the assessment to pass the module?

Yes

Methods for providing feedback on assessment

The assignment will be marked and feedback provided. Feedback will also be provided in class via the tutor demonstrating a solution to similar problems.

Assessment learning outcomes

	Game AI Implementation (100%)
Critically analyse the principles and techniques used in traditional game artificial intelligence, including pathfinding, decision systems, and behaviour modelling.	X
Evaluate and compare different AI approaches used in games, including rule-based systems and machine learning methods, and justify their suitability for specific design or technical problems.	X
Design and implement classical game AI algorithms within an interactive environment, such as pathfinding or behaviour systems for game agents.	X
Demonstrate a conceptual understanding of modern machine learning techniques, including neural networks, convolutional networks, recurrent networks, and transformer architectures.	X
Develop and apply neural network-based solutions to solve practical problems relevant to games or interactive systems.	X
Critically assess the role and impact of modern AI technologies in game development, including applications such as agent behaviour, content generation, rendering enhancement, and large language model integration.	X

Availability

Availability of module

Course	Route	Year of study	Type of availability	Required
Postgraduate Taught Games Engineering (Full-Time) (TWMS-H1SK)		1	Optional	No

Has pre-requisite modules?

No

Has post-requisite modules?

No

Has anti-requisite modules?

No

Any additional resource requirements

None.